

School of Computer Applications and Technology

Mid Term Examination - Apr 2026

M.Sc. CS II Sem / MCA (All) II Sem

Duration : 90 Minutes
Max Marks : 50

Sem II - E1PY210B - Machine Learning with Python

General Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

- | 1) | What are keywords and identifiers in Python? | CO1(3) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|------------|---|------------|-------------|----------------|--------------|----------------|--------------|-----|------|----|---|----|------|-----|--------|----|---|----|------|-----|------|----|---|----|------|-----|--------|----|---|----|------|-----|------|-----|---|----|------|-----|--------|----|---|----|------|-----|------|----|---|----|------|---------|
| 2) | Differentiate between Artificial Intelligence and Machine Learning with suitable examples. | CO2(5) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3) | Define Accuracy, Precision, Recall, and F1 Score with appropriate examples. | CO2(5) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4) | Write a Python program that creates a list of 10 integers, computes the maximum, minimum, and average values of the list, and explains the functions used to perform these operations. | CO1(8) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5) | Compare structured, semi-structured, and unstructured data in terms of format, storage, processing, and machine learning applications. | CO2(8) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6) | Given a confusion matrix with TP = 50, FP = 10, FN = 20, and TN = 120, calculate Accuracy, Precision, and Recall, and interpret the model's performance. | CO2(9) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7) | <p>A student performance dataset is given below:</p> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Student_ID</th> <th>Gender</th> <th>Attendance</th> <th>Study_Hours</th> <th>Previous_Marks</th> <th>Final_Result</th> </tr> </thead> <tbody> <tr><td>101</td><td>Male</td><td>85</td><td>3</td><td>78</td><td>Pass</td></tr> <tr><td>102</td><td>Female</td><td>85</td><td>4</td><td>82</td><td>Pass</td></tr> <tr><td>103</td><td>Male</td><td>92</td><td>3</td><td>88</td><td>Pass</td></tr> <tr><td>104</td><td>Female</td><td>76</td><td>5</td><td>78</td><td>Fail</td></tr> <tr><td>105</td><td>Male</td><td>100</td><td>2</td><td>60</td><td>Fail</td></tr> <tr><td>106</td><td>Female</td><td>88</td><td>3</td><td>77</td><td>Pass</td></tr> <tr><td>107</td><td>Male</td><td>85</td><td>3</td><td>78</td><td>Pass</td></tr> </tbody> </table> | Student_ID | Gender | Attendance | Study_Hours | Previous_Marks | Final_Result | 101 | Male | 85 | 3 | 78 | Pass | 102 | Female | 85 | 4 | 82 | Pass | 103 | Male | 92 | 3 | 88 | Pass | 104 | Female | 76 | 5 | 78 | Fail | 105 | Male | 100 | 2 | 60 | Fail | 106 | Female | 88 | 3 | 77 | Pass | 107 | Male | 85 | 3 | 78 | Pass | CO1(12) |
| Student_ID | Gender | Attendance | Study_Hours | Previous_Marks | Final_Result | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 101 | Male | 85 | 3 | 78 | Pass | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 102 | Female | 85 | 4 | 82 | Pass | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 103 | Male | 92 | 3 | 88 | Pass | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 104 | Female | 76 | 5 | 78 | Fail | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 105 | Male | 100 | 2 | 60 | Fail | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 106 | Female | 88 | 3 | 77 | Pass | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 107 | Male | 85 | 3 | 78 | Pass | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Perform the following tasks:

1. Identify and remove the unnecessary column that is not useful for Machine Learning processing.
2. Convert the categorical variables into numerical form using suitable encoding techniques.
3. Apply Min-Max Scaling on the numerical feature columns to bring all values into the range 0 to 1.
4. Prepare and display the final cleaned and preprocessed dataset after:

- Removing unnecessary columns
- Encoding categorical variables
- Applying feature scaling

5. Write the dataset in a form that is ready to be used for a Machine Learning model.

OR

Analyze the feature selection methods to reduce dimensionality in a dataset.

CO2(12)